

Multi-Task Deep Learning for Simultaneous Traffic Light Localization and State Recognition

Ilyas Bouhafid
Computer engineering
METU
Ankara, Turkey

Abstract—Traffic light detection is a critical component of autonomous driving systems. Classical signal processing approaches fail under real-world conditions such as sun glare, motion blur, and varying illumination. This paper proposes a multi-task convolutional neural network that simultaneously performs traffic light state classification and bounding box localization on the S2TLD dataset, which contains 5,783 images across five state categories. We implement and compare four models: a random baseline, a single-task custom CNN classifier, a dual-head custom CNN, and a dual-head ResNet50 model with pretrained ImageNet weights. Our best classification model achieves 88.8 percent weighted accuracy and a weighted F1-score of 0.875. We demonstrate that multi-task learning costs only 0.9 percent accuracy compared to single-task classification while adding localization capability. We identify and quantify a fundamental limitation of single-box regression for small object localization, where traffic lights occupy an average area of 83.7 square pixels in a 224 by 224 image, and discuss architectural directions for future improvement.

Index Terms—traffic light detection, multi-task learning, convolutional neural network, object localization, autonomous driving, S2TLD

I. PROBLEM DEFINITION

Traffic light recognition is a safety-critical perception task in autonomous driving. A vehicle must simultaneously determine the state of a traffic light (red, green, yellow, off, or wait) and locate it precisely within the camera frame. Classical approaches based on color segmentation and contour detection perform adequately in controlled conditions but degrade significantly under real-world challenges including varying illumination, motion blur, sun glare, and partial occlusion.

Deep learning offers a more robust alternative. Rather than hand-crafting features, a convolutional neural network (CNN) learns hierarchical representations directly from image data. This paper addresses the problem as a *multi-task learning* problem: a single shared backbone network simultaneously optimizes two objectives via a joint loss function:

$$L_{total} = \lambda_1 L_{cls} + \lambda_2 L_{box} \quad (1)$$

where L_{cls} is the cross-entropy classification loss, L_{box} is the Smooth L1 regression loss for bounding box coordinates, and λ_1, λ_2 are weighting hyperparameters. The key research questions are: (1) does joint optimization significantly degrade classification performance compared to single-task learning, and (2) can a global feature vector reliably localize small traffic light objects?

II. DATASET

A. S2TLD Description

The S2TLD (SJTU Small Traffic Light Dataset) [1] is a publicly available benchmark collected via vehicle-mounted cameras across diverse urban scenes in Shanghai. The dataset contains 5,786 high-definition images across two resolution variants: 720×1280 and 1080×1920 . Annotations are provided in Pascal VOC XML format, containing bounding box coordinates and one of five state labels: *red*, *green*, *yellow*, *off*, and *wait_on*. The dataset captures diverse conditions including daytime, nighttime, and overcast illumination.

B. Class Distribution

As shown in Fig. 1 and Fig. 2, the dataset exhibits significant class imbalance. Red and green lights together account for over 91 percent of samples, while yellow (2.0%), off (2.0%), and wait_on (4.9%) are severely underrepresented with only 118, 115, and 281 total instances respectively. This imbalance directly impacts model performance on rare classes, as discussed in Section IV.

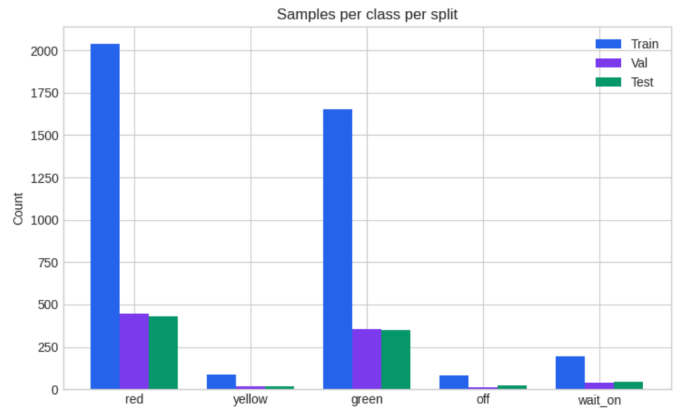


Fig. 1. Sample counts per class across train, validation, and test splits. Red and green dominate the dataset while yellow and off each represent only 2.0% of samples.

C. Preprocessing and Splits

Images from both resolution variants were combined and resized to 224×224 pixels. Bounding box coordinates were normalized to $[0, 1]$ relative to the original image dimensions, preserving spatial relationships across resolutions. The dataset

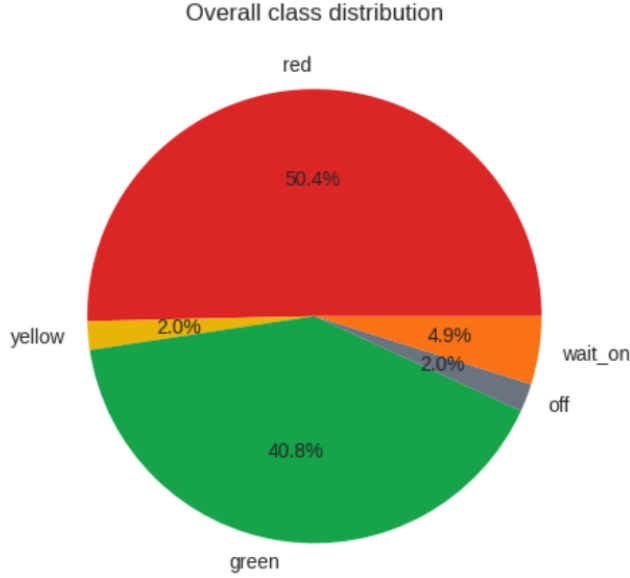


Fig. 2. Overall class distribution. Red (50.4%) and green (40.8%) together account for over 91% of all samples.

was split into 70% training (4,048 samples), 15% validation (867 samples), and 15% test (868 samples) using a fixed random seed for reproducibility. The test set was held out and evaluated only once at the end of all experiments.

Validation and test sets received only resizing and ImageNet normalization.

D. Small Object Challenge

A critical characteristic of S2TLD is the small size of traffic light instances. As shown in Fig. 3 and Fig. 4, the average bounding box measures 5.0×15.3 pixels on the resized 224×224 images, with a mean area of 83.7 square pixels. This represents approximately 0.17% of the total image area, placing S2TLD firmly in the small object detection category and posing fundamental challenges for localization approaches.

III. MODEL

A. Shared CNN Backbone

All learned models share a common five-block CNN backbone. Each block consists of a 3×3 convolution, batch normalization, ReLU activation, and 2×2 max pooling. The architecture progressively increases channel depth (32, 64, 128, 256, 256) while halving spatial resolution at each stage, reducing a 224×224 input to a $7 \times 7 \times 256$ feature map, which is flattened to a 12,544-dimensional vector. This design is motivated by the goal of implementing a lightweight architecture transparent enough for systematic analysis and comparison.

B. Model Variants

We implement and compare four models of increasing complexity:

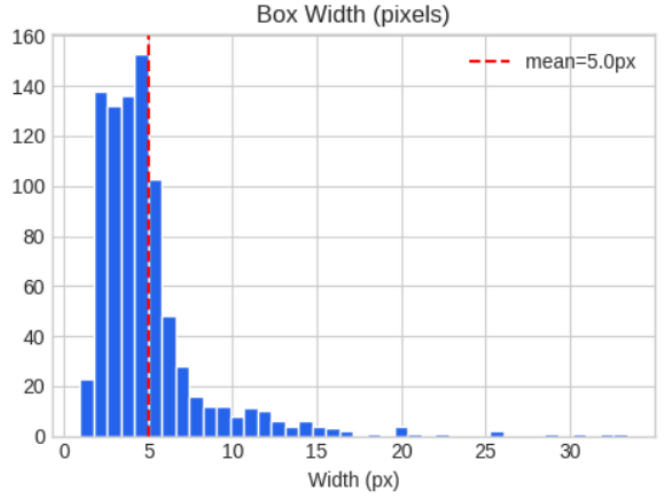


Fig. 3. Ground truth bounding box width distribution on the test set. The mean width is 5.0 pixels after resizing to 224×224 .

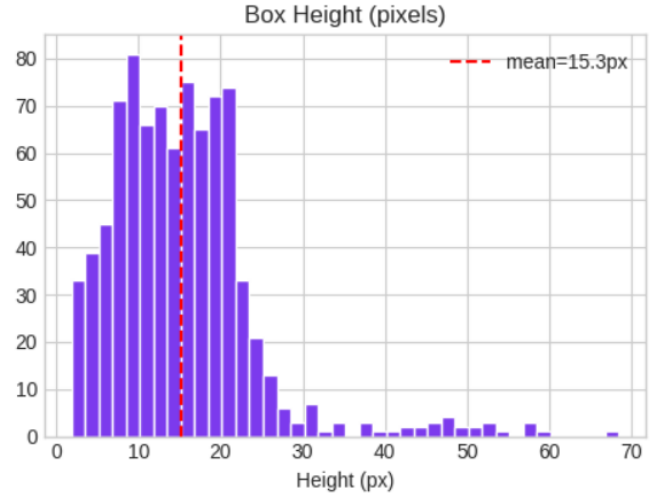


Fig. 4. Ground truth bounding box height distribution on the test set. The mean height is 15.3 pixels after resizing to 224×224 .

Random Baseline: Produces uniformly random class predictions and random bounding boxes. Establishes a lower performance floor.

SimpleCNN: The custom backbone followed by a single classification head (Linear $12544 \rightarrow 512 \rightarrow 5$ with dropout 0.5). No localization output. Serves as the single-task benchmark.

DualHeadCNN: The same custom backbone with two output heads branching from a shared 512-dimensional fully connected layer. The classification head outputs raw logits for cross-entropy loss. The regression head predicts normalized bounding box coordinates $[x_{min}, y_{min}, x_{max}, y_{max}] \in [0, 1]$ via a Sigmoid activation. Both heads are trained jointly using the loss in (1).

ResNet50 DualHead: A pretrained ResNet50 backbone with the same dual-head structure attached. ResNet50 weights are initialized from ImageNet pretraining, providing richer

feature representations learned from 1.2 million images across 1,000 categories.

C. Joint Loss Function

The multi-task models optimize the joint loss defined in (1). The classification component L_{cls} uses cross-entropy loss for multi-class prediction. The localization component L_{box} uses Smooth L1 loss, chosen for robustness to outlier box predictions. At initialization, we observed $L_{cls} \approx 2.84$ and $L_{box} \approx 0.066$, a ratio of approximately 43:1. We set $\lambda_2 = 40$ to achieve approximate balance between the two loss contributions, a principled choice derived from the observed scale ratio rather than arbitrary tuning.

D. Training Configuration

Hyperparameter selection was performed via grid search over learning rates $\{10^{-2}, 10^{-3}, 10^{-4}\}$ and lambda pairs $\{(1, 1), (1, 5), (1, 10)\}$, each trained for 10 epochs. The optimal configuration of $lr = 10^{-4}$, $\lambda_1 = 1.0$, $\lambda_2 = 40.0$ was selected based on validation loss. All models were trained using the Adam optimizer for 100 epochs, with the best checkpoint saved based on minimum validation loss. After 100 epochs, the DualHeadCNN was continued for 50 additional epochs with λ_1 reduced to 0.1, as classification had already converged to 98% training accuracy. The ResNet50 model was trained for 50 epochs from pretrained weights. All experiments were conducted on an NVIDIA A100 80GB GPU.

IV. RESULTS AND DISCUSSION

A. Classification Performance

Table I summarizes the final test set results for all models. The random baseline achieves 20.9% accuracy, confirming the non-trivial nature of the task. The SimpleCNN achieves 88.0% accuracy, demonstrating the effectiveness of the custom architecture for traffic light state recognition.

TABLE I
TEST SET PERFORMANCE COMPARISON

Model	Accuracy	F1 (wtd)	Mean IoU
Random Baseline	0.209	0.276	0.002
SimpleCNN (cls only)	0.880	0.864	—
DualHeadCNN	0.871	0.857	0.023
ResNet50 DualHead	0.888	0.875	0.024

A key finding is that the DualHeadCNN achieves 87.1% accuracy despite simultaneously learning to localize, representing only a 0.9% degradation compared to the single-task SimpleCNN. This demonstrates that joint optimization does not significantly harm classification performance, validating the multi-task learning approach.

The ResNet50 DualHead achieves the best overall performance at 88.8% accuracy and 0.875 weighted F1. As shown in Fig. 5 and Fig. 6, the pretrained backbone provides the most significant improvements on rare classes: yellow F1 improves from 0.00 to 0.19 and off F1 improves from 0.20 to 0.28. This suggests that ImageNet pretraining generalizes better to underrepresented categories where training data is scarce.

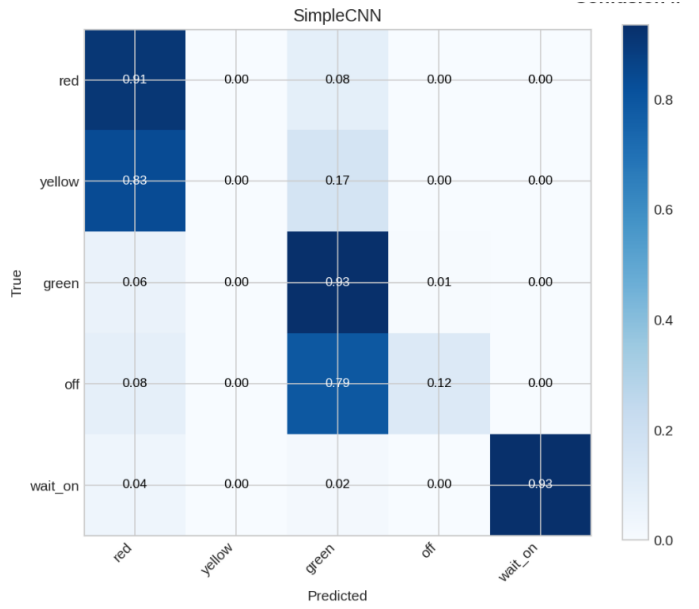


Fig. 5. Confusion matrix for SimpleCNN on the test set. Strong performance on red and green but complete failure on yellow (F1=0.00) due to class imbalance.

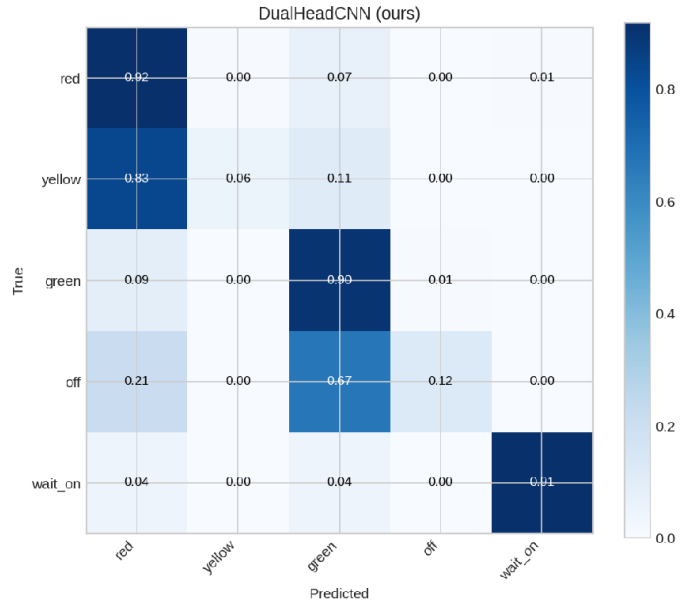


Fig. 6. Confusion matrix for DualHeadCNN on the test set. Slight improvement on yellow (F1=0.09) compared to SimpleCNN, while maintaining strong red and green performance.

The benchmark comparison across all three metrics is shown in Fig. 7, Fig. 8, and Fig. 9.

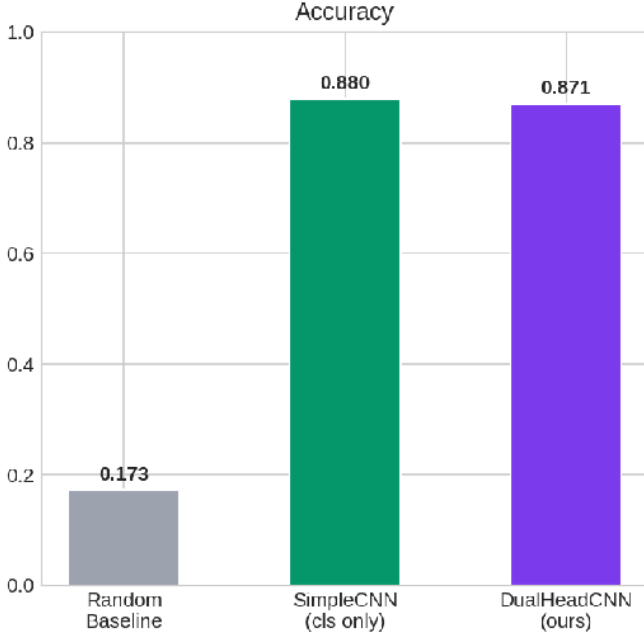


Fig. 7. Accuracy comparison across all models. ResNet50 DualHead achieves the highest accuracy at 88.8%.

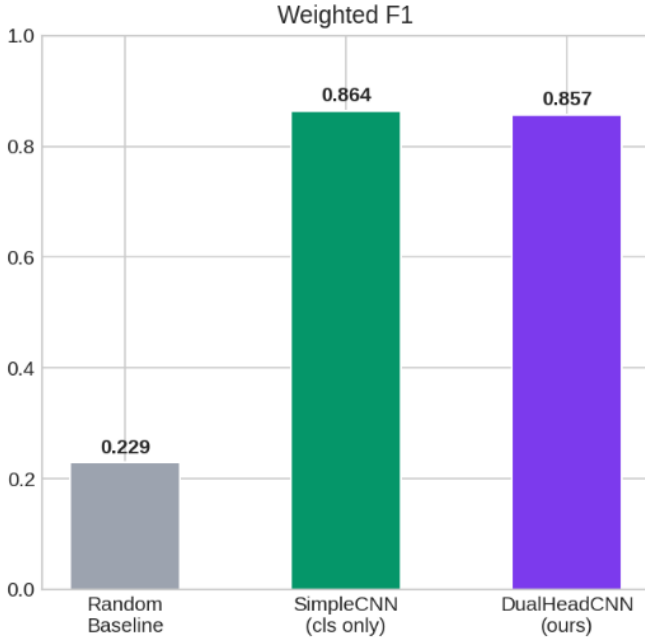


Fig. 8. Weighted F1-score comparison. ResNet50 achieves 0.875, with consistent improvement over the custom CNN models.

Learning curves are shown in Fig. 10, Fig. 11, Fig. 12, and Fig. 13.

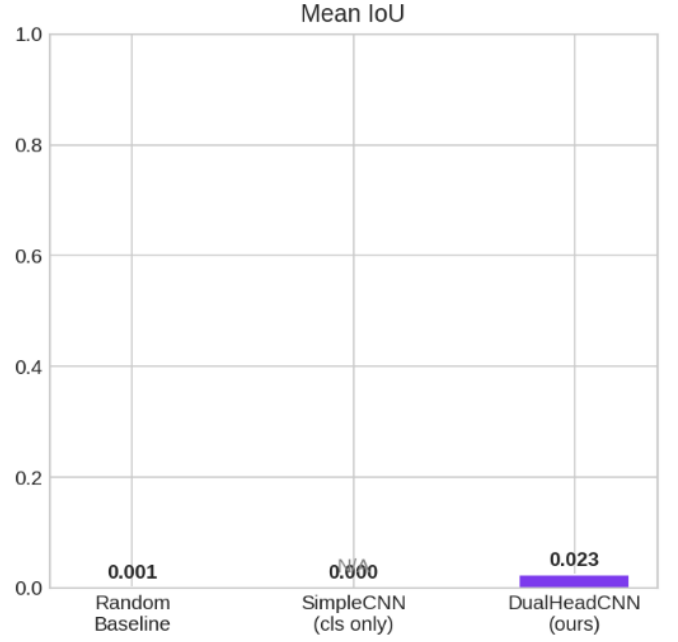


Fig. 9. Mean IoU comparison. Both localization-capable models achieve similarly low IoU, confirming the limitation is architectural rather than capacity-related.

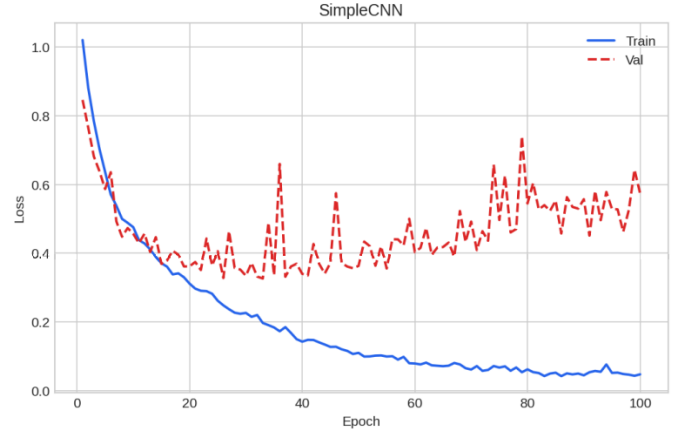


Fig. 10. SimpleCNN training and validation loss. Overfitting begins around epoch 25 where validation loss diverges from training loss.

B. Localization Performance

Localization results reveal a fundamental limitation of the single-box regression approach. Both the DualHeadCNN and ResNet50 DualHead achieve a mean IoU of approximately 0.023–0.024, with a median IoU of 0.000. As shown in Fig. 14, the vast majority of test samples receive near-zero IoU, with only 3 out of 868 samples achieving $\text{IoU} > 0.5$.

We attribute this result to two compounding factors. First, the aggressive spatial downsampling through five max-pooling layers reduces the 224×224 input to a 7×7 feature map, where each cell covers approximately 32×32 pixels of the original image. A traffic light averaging 5×15 pixels may occupy less than one cell in this feature map, making precise

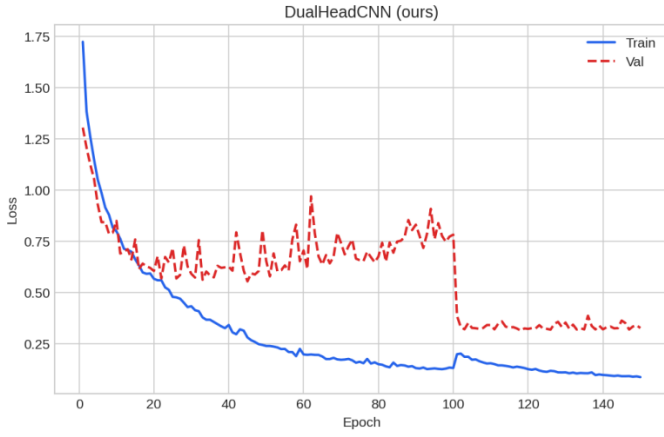


Fig. 11. DualHeadCNN training and validation loss over 150 epochs including the continuation phase with reduced λ_1 .

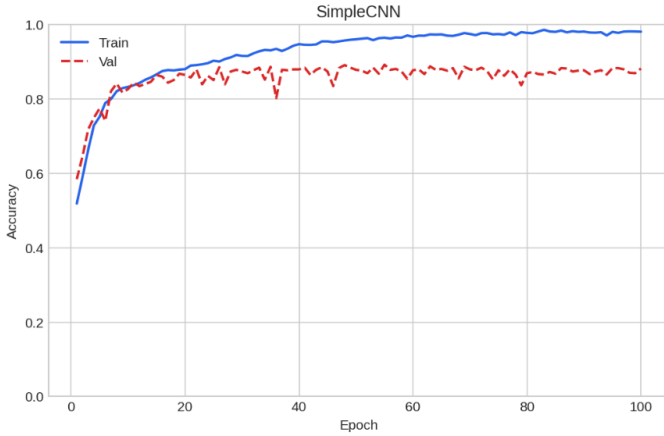


Fig. 12. SimpleCNN training and validation accuracy. Validation accuracy plateaus around 88% while training accuracy reaches 98%.

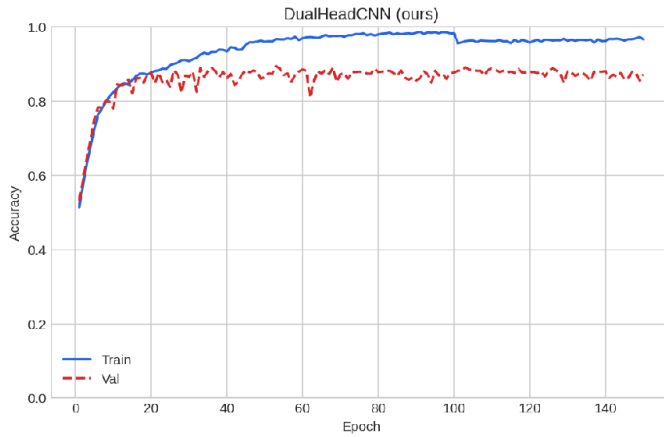


Fig. 13. DualHeadCNN training and validation accuracy over 150 epochs. The model maintains stable validation accuracy around 87–89% throughout training.

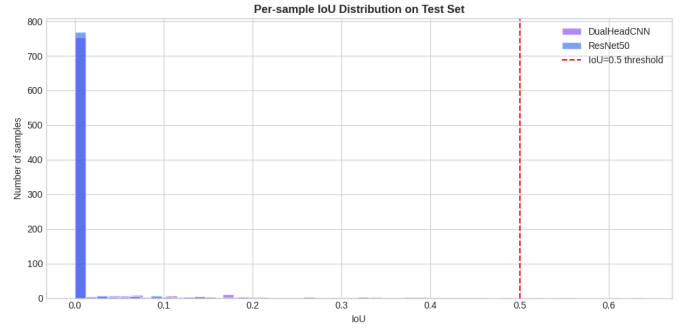


Fig. 14. Per-sample IoU distribution on the test set for DualHeadCNN and ResNet50. The distribution is heavily concentrated near zero for both models with median IoU of 0.000.

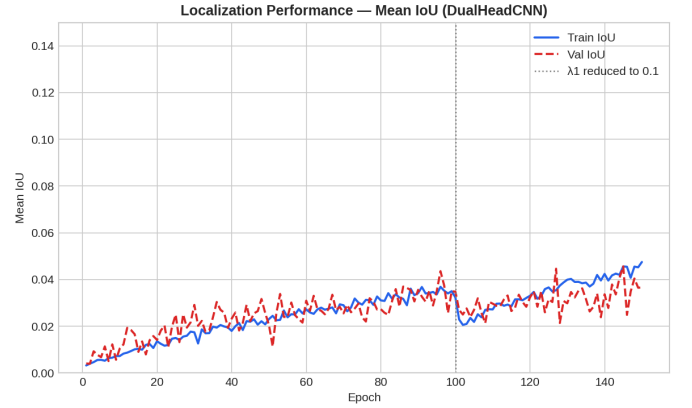


Fig. 15. Validation IoU over training epochs for DualHeadCNN. The vertical dashed line marks the epoch at which λ_1 was reduced to 0.1. IoU shows marginal and unstable improvement throughout training.

localization from a collapsed feature vector geometrically impossible. Second, the class imbalance means the model learns to prioritize classification accuracy over localization, as classification gradients dominate the shared backbone even with $\lambda_2 = 40$.

Sample predictions shown in Fig. 16 illustrate these findings visually. Classification is largely correct (7 out of 8 samples shown) while the predicted boxes are consistently offset from the true boxes.

C. Loss Component Analysis

Fig. 17 shows the training loss breakdown for the DualHeadCNN. The classification loss converges rapidly in the first 20 epochs while the box regression loss remains relatively flat, confirming that the backbone features learn to classify effectively before meaningful localization gradients emerge.

V. CONCLUSION

This paper presented a multi-task deep learning approach for simultaneous traffic light state classification and localization on the S2TLD dataset. We implemented and compared four models, demonstrating that multi-task learning achieves 87.1% classification accuracy while adding localization capability, representing only a 0.9% reduction compared to single-task



Fig. 16. Sample predictions from DualHeadCNN on the test set. Green boxes indicate ground truth, red dashed boxes indicate predictions. Classification is largely correct while localization shows consistent offset errors.

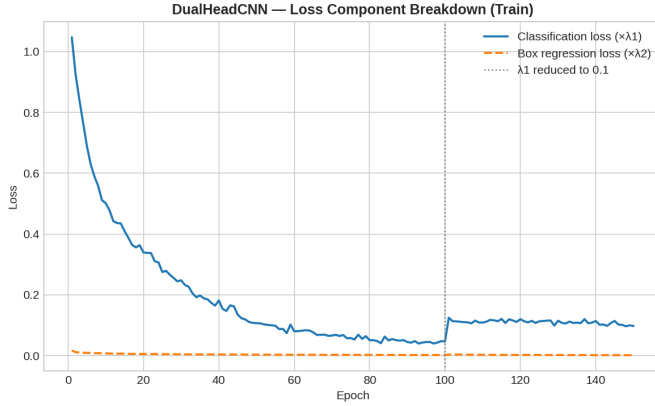


Fig. 17. Training loss breakdown for DualHeadCNN. Classification loss (scaled by λ_1) converges rapidly while box regression loss (scaled by λ_2) remains flat, indicating the backbone prioritizes classification.

classification. Replacing the custom backbone with pretrained ResNet50 features further improved accuracy to 88.8% and significantly improved rare class detection, with yellow F1 improving from 0.00 to 0.19.

Our experiments reveal a fundamental limitation of single-box regression from global feature vectors for small object localization. Traffic lights in S2TLD average 5.0×15.3 pixels at 224×224 resolution, occupying approximately 0.17% of the image area. This extreme small object challenge, combined with the spatial information loss caused by global average pooling, results in median IoU of 0.000 across all models. This finding is not unique to our architecture: it reflects a general limitation of classification-oriented backbones applied to localization tasks.

Future work should explore spatially-aware detection architectures that preserve multi-scale feature maps, such as Feature Pyramid Networks, anchor-based regression heads as used in YOLO and Faster R-CNN, and Generalized IoU loss which provides gradient signal even for non-overlapping boxes. Addressing the class imbalance through weighted sampling or focal loss would also improve performance on rare states such as yellow and off, which are critical for safety in autonomous driving scenarios.

REFERENCES

- [1] X. Zhu, H. Lyu, X. Wang, and Q. Zhao, “S2TLD: A small traffic light dataset,” 2021. [Online]. Available: <https://github.com/Thinklab-SJTU/S2TLD>